

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) State and explain Kirchhoff's current and voltage laws with examples. 4
- b) For the network shown in Fig. 1.1, calculate the resistance R_{ab} and the current i . Also, calculate the voltage drop across 13Ω resistor. 6

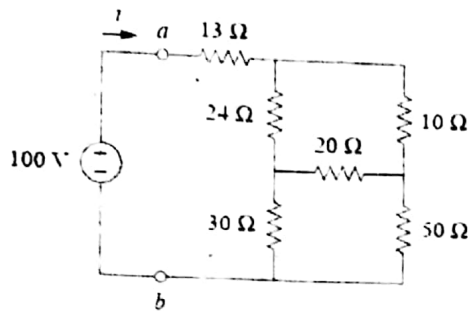


Fig. 1.1

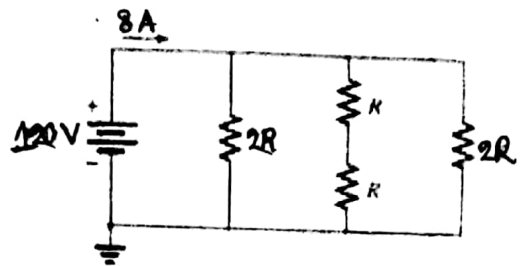


Fig. 1.2

- c) In the circuit in Fig. 1.2, what is the value of the resistors used? 4

2. a) For Wye-Delta conversion Prove $R_\Delta = 3R_Y$ 4
- b) For the following network in Fig. 2.1 Find R_{eq} and i_o 5
- c) Find node voltages using nodal analysis of the following circuit in Fig. 2.2. 5

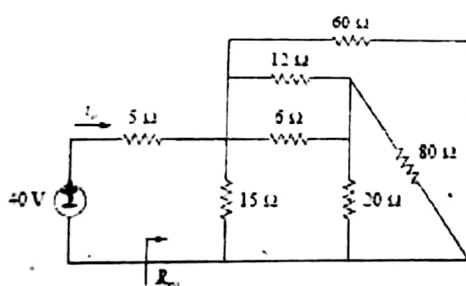


Fig. 2.1

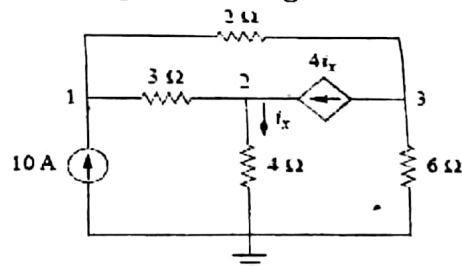


Fig. 2.2

3. a) What do you mean by super node and super mesh? 2
- b) Find V_1 , V_2 , V_3 of the following circuit in Fig. 3.1. 6
- c) Using mesh analysis determine I_1 , I_2 , I_3 of the following circuit in Fig. 3.2. 6

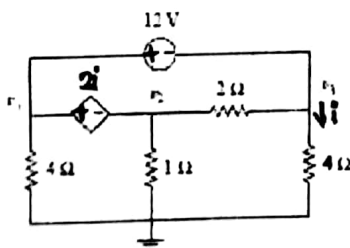


Fig. 3.1

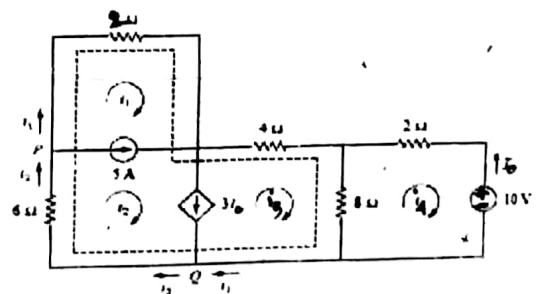


Fig. 3.2

4. a) Use Thevenin's theorem to find the equivalent circuit to the left of the terminals a and b shown in Fig. 4.1. Also, calculate the value of I . 5
 b) Use source transformation method to calculate v_0 in the circuit in Fig. 4.2. 4

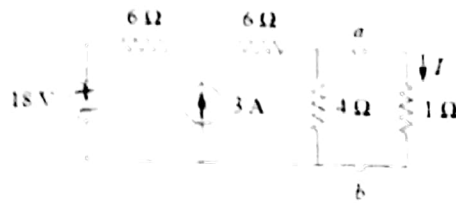


Fig. 4.1

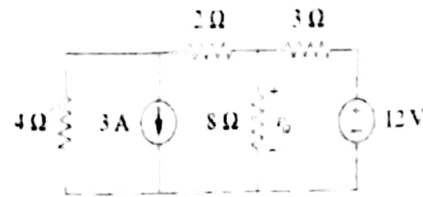


Fig. 4.2

5. c) Find Norton's equivalent circuit of the following circuit in Fig. 4.3. 5

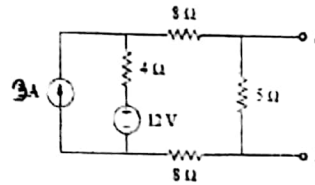


Fig. 4.3

5. a) Find the condition for maximum power and derive the equation for maximum power. 4
 b) Find Thevenin equivalent circuit of the following circuit in Fig. 5.1. 5
 c) Find I_0 of the circuit in Fig 5.2 using source transformation. 5

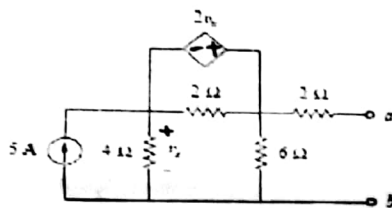


Fig. 5.1

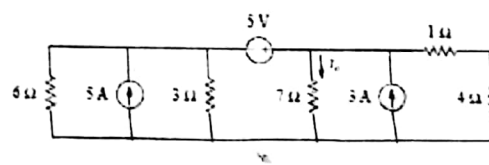


Fig. 5.2

6. a) State and explain Faraday's law of Electromagnetic Induction. 4
 b) Derive the equation for energy stored in an inductor. 4
 c) Determine the input impedance for the circuit in Fig. 6.1. Assume $\omega = 10 \text{ rad/s}$. 6

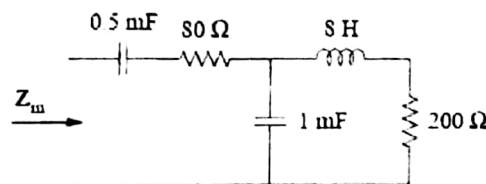


Fig. 6.1

7. a) Derive the equation of average power absorbed by a resistor. 4
 b) Determine the rms value of the current waveform of the following Fig. 7.1. If the current pass through a $2\text{-}\Omega$ resistor, find the average power absorbed by the resistor. 4
 c) Determine the average power generated by each source and average power absorbed by each passive element of the following circuit in Fig. 7.2. 6

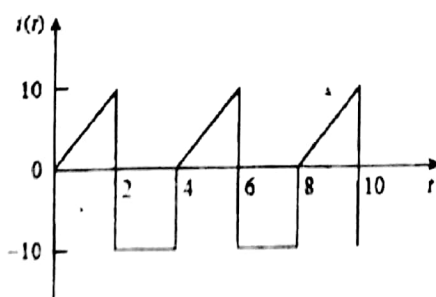


Fig. 7.1

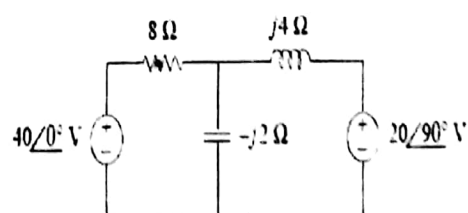


Fig. 7.2